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Editor-in-Chief
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Dear Editor,

I am pleased to submit my manuscript, "A Near-Linear-Time Solver for Bilevel Congestion Network Design via Directed Nonlinear Laplacian Flow," for consideration as a Research Article in INFORMS Journal on Computing.

The problem. Bilevel network design—computing tolls, capacity expansions, or routing incentives that minimize total system travel time subject to a user-equilibrium response—has been understood at the level of sensitivity analysis since Tobin and Friesz (1988): the design gradient reduces to one linearized-equilibrium solve per Newton step, independent of the number of design variables. In principle this makes gradient-based bilevel design exact and scalable. In practice it has not been, because that sensitivity solve is itself a direct factorization of the equilibrium Jacobian, and direct factorization grows superlinearly on the irregular, poorly-separable networks that selfish-routing and congestion-pricing applications actually present—road networks with hub-like structure, communication and overlay networks, peer-to-peer topologies. The result is a one-to-two-order-of-magnitude gap between how large a network we can *solve* for equilibrium and how large a network we can *design* for, even though the underlying mathematics says the two should cost about the same.

The bridge. This paper closes that gap by recognizing that the directed user equilibrium, written as a source-form nonlinear Laplacian flow, has a Newton Jacobian that remains a symmetric weighted graph Laplacian on the active subgraph—despite the one-sided, rectified congestion law that traffic and communication routing actually obeys. That one structural fact is the bridge between the transportation application and the algorithmic core: it means every solve the bilevel design loop needs—the forward equilibrium *and* the design adjoint—reduces to a short sequence of near-linear-time Laplacian solves, using the same class of combinatorial multigrid and approximate-Cholesky solvers that have transformed large-scale spectral graph algorithms over the last decade. A smoothing homotopy in the dead-zone width handles the one complication (the rectified law's ill-conditioning and shifting active set) without sacrificing that near-linear cost.

Why I believe this belongs in IJOC. The contribution is squarely at the intersection IJOC serves: a computational bottleneck in a classical OR problem (bilevel congestion pricing / network design), diagnosed precisely as a linear-algebra scaling issue, and resolved with near-linear numerical methods, validated with the rigor a computing-focused readership expects rather than asserted from theory alone:

- The forward-solve scaling is confirmed on a topology-diverse corpus of 96 real irregular graphs spanning four orders of magnitude (1.1×10^4 to 6.0×10^6 arcs) drawn from SuiteSparse/SNAP, LAW, DIMACS10, and Newman, not just a single synthetic family.
- The comparison against direct factorization is a fair fight, not a strawman: alongside the standard unsymmetric UMFPACK/COLAMD baseline, I benchmark against CHOLMOD under a METIS nested-dissection ordering—the theoretically appropriate direct solver for the symmetric positive-definite Jacobian this problem produces—and show the near-linear solver still wins with a widening margin, and completes at network sizes where even that stronger direct solver exhausts memory.
- The framework extends to multicommodity logit stochastic user equilibrium, validated on the Sioux Falls and Anaheim benchmark networks with their published origin–destination matrices, where destination-bundled toll design reduces total system travel time by 7.7–13.1%.
- Every experiment is reproducible: code, unit tests, and reproduction scripts are publicly available at https://github.com/orenlivne/dnlf, and I have prepared the accompanying software/data archive per IJOC's reproducibility requirements.

We are candid about scope: the advantage is specific to poorly-separable, non-planar topologies—on near-planar road networks, where nested dissection is already near-optimal, the method does not win. I believe this framing, together with the depth of the empirical validation, makes the paper a good fit for IJOC's standards. Thank you for your consideration.

Sincerely,
Oren E. Livne